Getting your laptop ready

Quantitative Methods in Neuroscience, Master in Neuroscience, University of Geneva, 2026-27

The practicals in this course are Python notebooks that you run on your own laptop. This guide takes you from a blank machine to a working setup. Please do it **before the first practical**, so that we can spend that session on the material rather than on installations. You do it once and never again, and no previous experience with Python or with a terminal is assumed.

About 30 min
Most of it waiting for downloads

5 GB free
Environment, installers and cache

No experience
Nothing to know beforehand

What you will end up with

conda
The tool that manages Python and the libraries the course uses.

Visual Studio Code
The editor in which you open, write and run the notebooks.

The `qmn` environment
One place holding every library the course needs, kept apart from the rest of your machine.

What you need before you start

- A reliable internet connection, since nearly everything here is a download.
- The file **environment.yml**, attached to the same message as this guide. Save it somewhere you can find again, you will need it in Step 3.
- Half an hour in which the laptop can stay on.

The six steps

1. **Install conda**, which brings a Python along with it.
2. **Install VS Code**, plus its Python and Jupyter extensions.
3. **Put the course folder together** from the files on Moodle.
4. **Build the qmn environment**, one command that installs every library.
5. **Check that it worked**, one line that should print **all good**.
6. **Choose the environment inside a notebook**, which is the kernel.

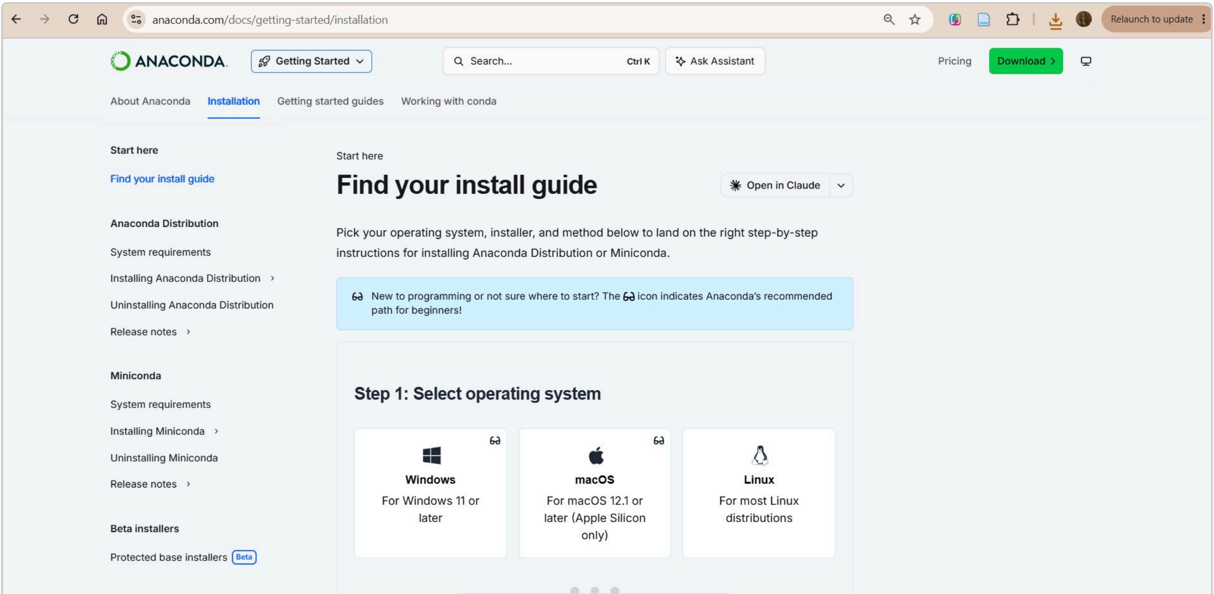
> Do the steps in order, and read each one through before starting it. If something goes wrong on the way, the last page says what to do.

Step 1. Install conda

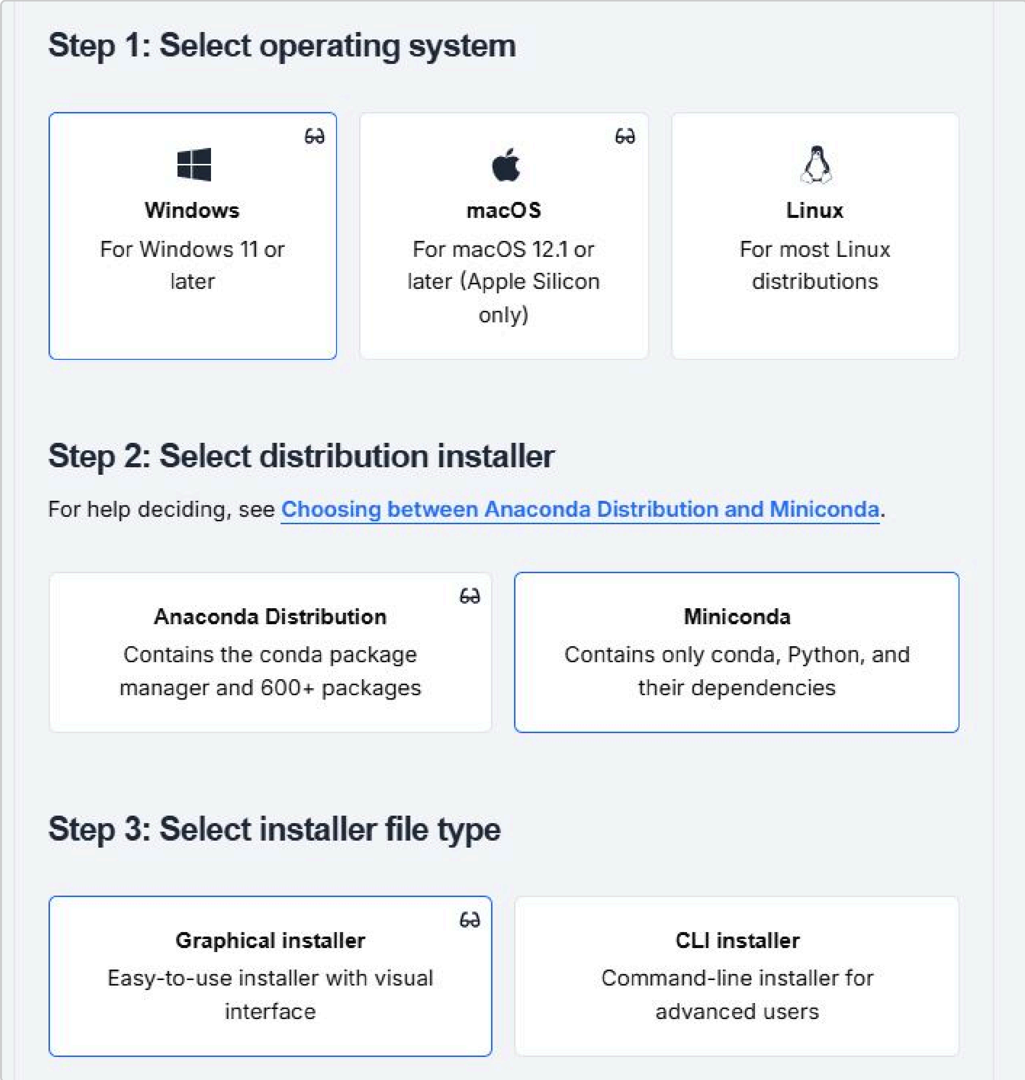
> **What conda is, and why we use an environment.** Conda installs Python and the scientific libraries that come with it, and it can hold several sets of them on the same machine without them interfering. One such set is called an environment. Ours is named `qmn`: it holds every library the practicals need, at the versions we choose, and nothing outside it is touched. That is what makes the notebooks behave the same way on every laptop in the class. It also means that if the environment ever ends up in a bad state, we rebuild it with one command and the rest of your machine is unaffected.

Go to the Anaconda installation page and pick, in order, your operating system, then **Miniconda**, then the **graphical installer**:

```
https://www.anaconda.com/docs/getting-started/miniconda/install
```



The page opens on "Find your install guide". Start with Step 1 and choose your system.



Choose **Miniconda** in Step 2 and the **graphical installer** in Step 3, then follow the instructions the page gives you for your system.

Run the installer and proceed with the options it recommends. On Windows, leave the PATH option as the installer sets it.

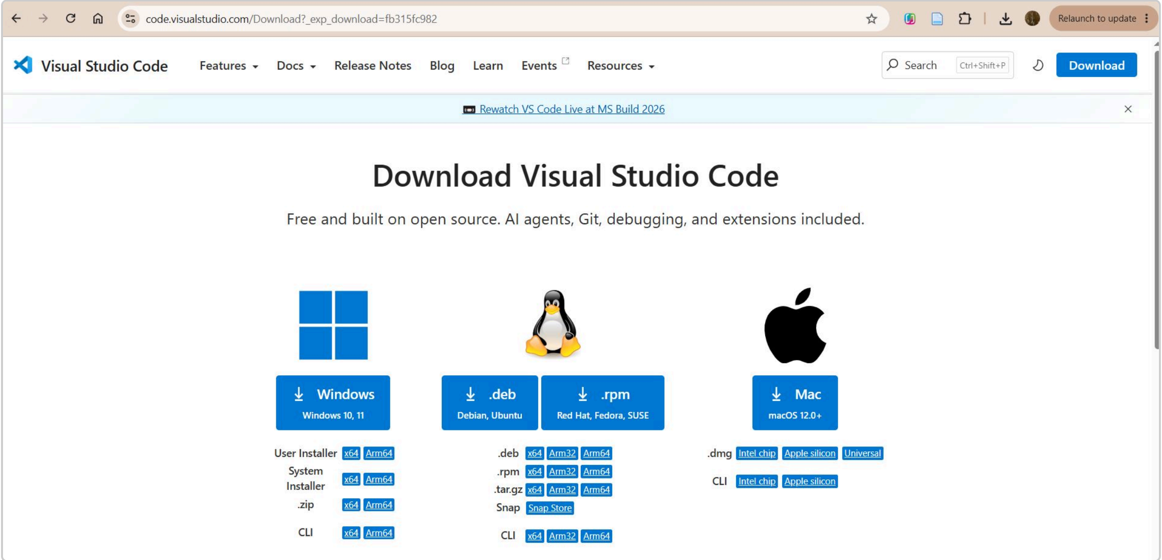
> **Already have Anaconda?** Then you already have conda and can skip this step. Anaconda and Miniconda are the same tool; Miniconda simply installs fewer things you will not use.

Step 2. Install Visual Studio Code

> **What VS Code is.** An editor. It opens the notebook files, runs the code in them one cell at a time, and shows the plots next to the code that made them. It also opens plain Python files, the ones ending in `.py`, which is how code is written outside a notebook. It is free and behaves the same on Windows, macOS and Linux.

Download it for your system and install it with the default options:

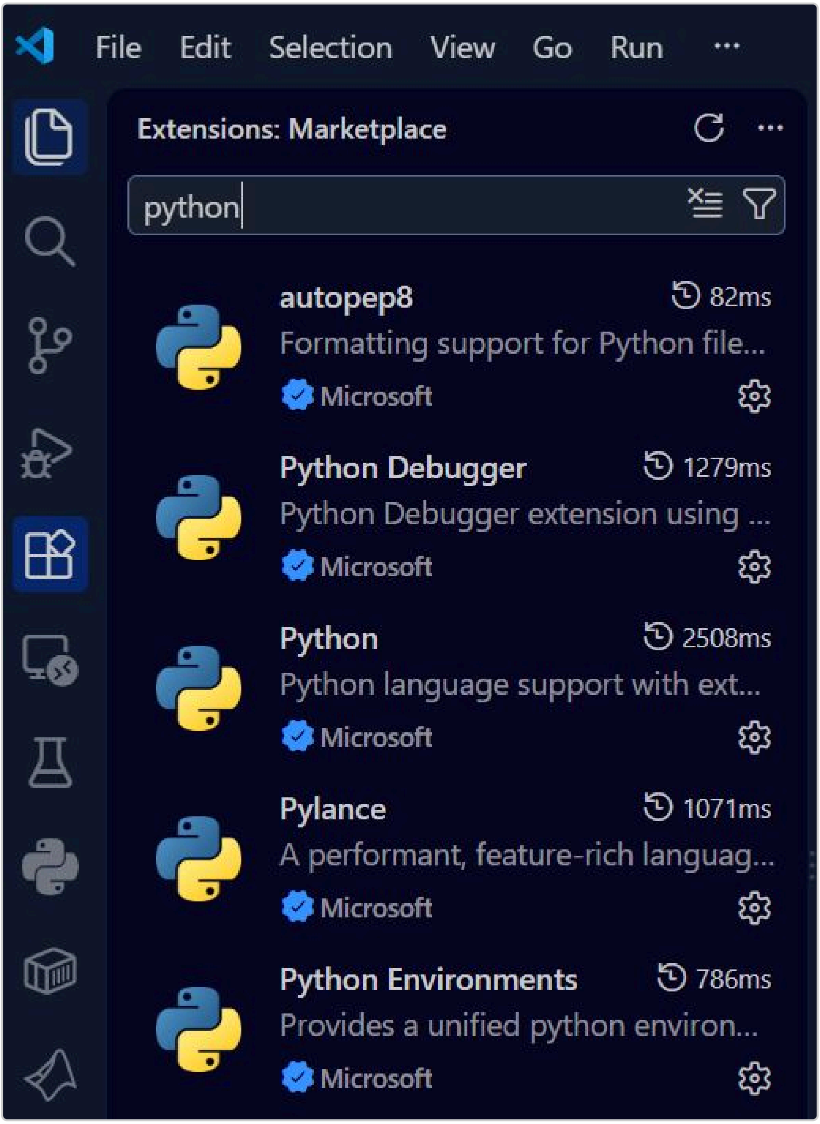
```
https://code.visualstudio.com/Download
```



Pick the button for your operating system. On a Mac, choose the version matching your chip if you are asked, Apple silicon or Intel.

Then open VS Code and click the **Extensions** icon in the left bar, the one made of four small squares. Search for these two by name and install them:

- **Python**, which also brings Pylance and the Python debugger
- **Jupyter**, which lets you run notebooks inside VS Code



Searching brings back several results, all published by Microsoft. Install the one named exactly **Python**, then search again for **Jupyter** and install the one named exactly that. The others arrive on their own or are not needed.

Step 3. Put the course folder together

> If you received everything as a single zip file rather than as separate downloads, this step is already done for you: unzip it, and go straight to Step 4.

The material is on Moodle as several separate files, so that a corrected notebook can be replaced later without you downloading everything again. You put them together once, into a single folder, and that folder is where you work for the rest of the semester.

1. Download **QMN_support_files.zip** and unzip it directly inside your user folder, the one named after you. It makes a folder called **QMN_practicals** holding the data, the images and a few helper files.
2. Download **environment.yml** and put it in that same folder. It is a short text file listing the libraries the course uses.
3. Download each **notebook** and put it in that same folder as well, next to `data`. Do the same for every notebook posted later in the semester.

```
Windows       %USERPROFILE%\QMN_practicals
macOS, Linux  ~/QMN_practicals
```

`%USERPROFILE%` and ~ are shorthand for that folder, so you never have to type your user name. Another place works too, inside Documents for instance, as long as you use your own path in the commands of Step 4. When you are done the folder looks like this:

```
QMN_practicals
    data                                  from the zip
    assets                                from the zip
    src                                   from the zip
    environment.yml
    01_programming_fundamentals.ipynb     one Moodle item each
    02_math_refresher.ipynb
    03_probability_descriptive.ipynb
    04_inference_ttests_power.ipynb
```

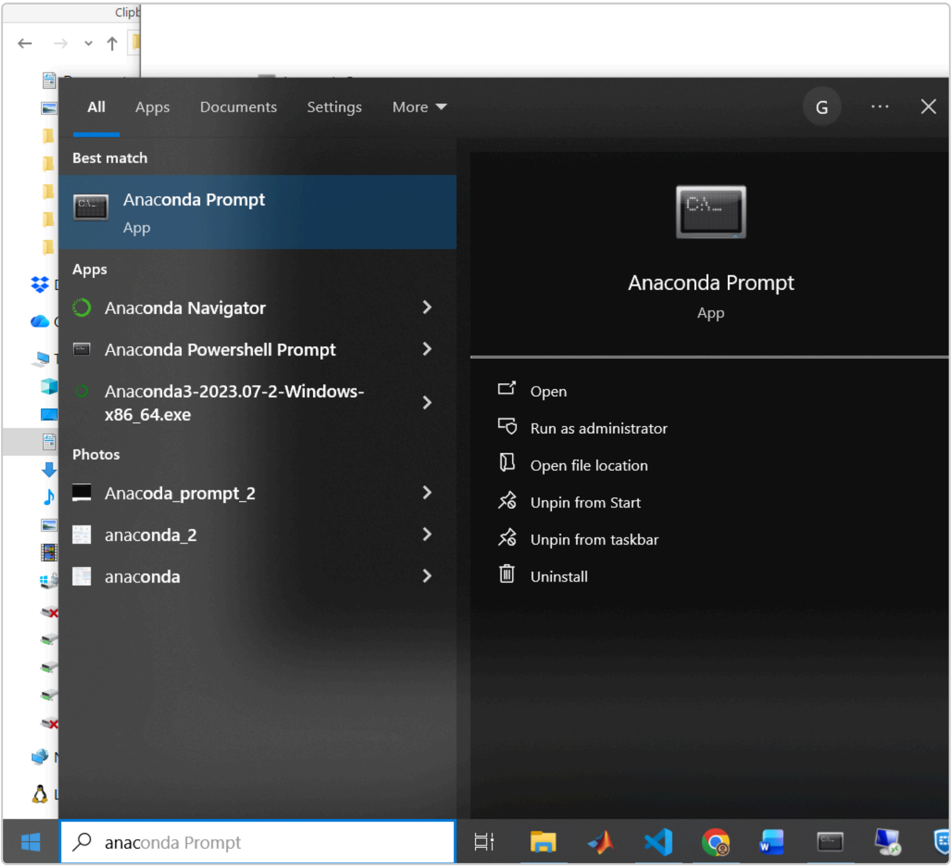
> **This is the part that goes wrong.** A notebook left in your Downloads folder cannot find the data, and its first cell stops with an error about a missing file. Each notebook looks for `data` in the folder it is sitting in, so it has to sit next to it. The cheat sheets and this guide are yours to keep anywhere: nothing reads them.

> **Check the name of the file.** It has to be exactly `environment.yml`. Windows hides file extensions by default, so a file that looks right in the folder can in fact be `environment.yml.txt`, and a second download usually arrives as `environment (1).yml`. In File Explorer turn on **View**, then **Show**, then **File name extensions** to see the real name, and rename the file if it needs it.

Step 4. Create the qmn environment

Open a terminal in which conda is available:

- **Windows**: press the Windows key, type `anaconda prompt`, and open the **Anaconda Prompt** app
- **macOS or Linux**: open the Terminal application



On Windows, look for "Anaconda Prompt" in the start menu rather than using the normal command prompt.

Check that conda is there. Type this and press Enter:

```
conda --version
```

You should see something like `conda 24.9.0`. If instead you get "command not found", close the window, open it again, and try once more.

Now move into the folder you made in Step 3 and list what is inside it:

```
cd %USERPROFILE%\QMN_practicals
dir
```

On macOS and Linux these two are `cd ~/QMN_practicals` and `ls`. You should see **environment.yml** in the list. If you do not, you are in a different folder or the file is not named the way you think, and it is worth sorting that out before going further.

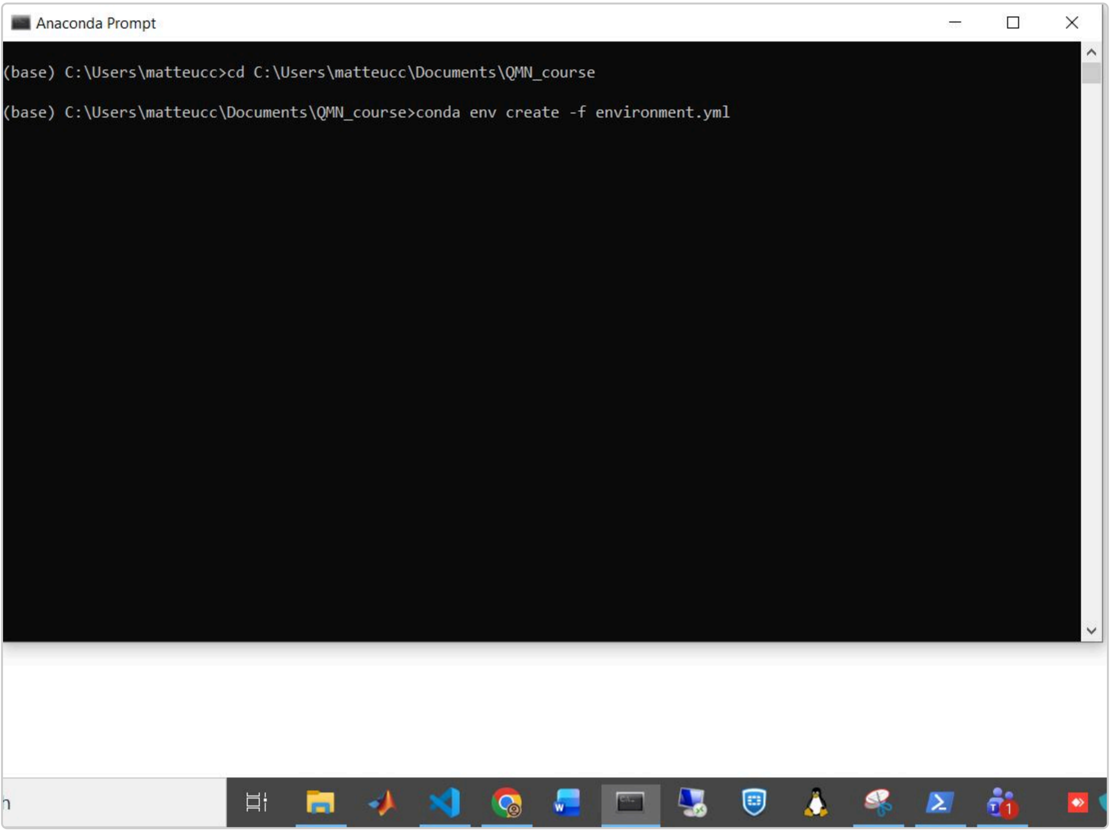
Then create the environment. This command reads `environment.yml` and installs everything it lists:

```
conda env create -f environment.yml
```

> **If conda asks you to accept Terms of Service**, and it may, it stops and prints a command beginning with `conda tos accept`. Copy the whole line it gives you, run it, and then run the create command above again. This is Anaconda asking you to agree to their terms rather than anything being broken, and it happens once per machine.

> **If conda tells you that `qmn` already exists**, you are doing this a second time, on a machine where the environment was already set up once. Do not delete it and do not run the create command again. Bring the one you have up to date instead, from the same folder:

```
conda env update -n qmn -f environment.yml --prune
```



Changing folder, then creating the environment. On this machine the course folder sits inside Documents, which is why the path is longer than the one above.

This downloads and installs Python and every library the course needs, so it takes roughly 5 to 15 minutes depending on your connection. It is normal for it to look stuck for a while on "Solving environment". When it finishes, activate the environment:

```
conda activate qmn
```

Your prompt should now start with `(qmn)`.

Step 5. Check that it worked

With `(qmn)` showing in the prompt, type this one line and press Enter:

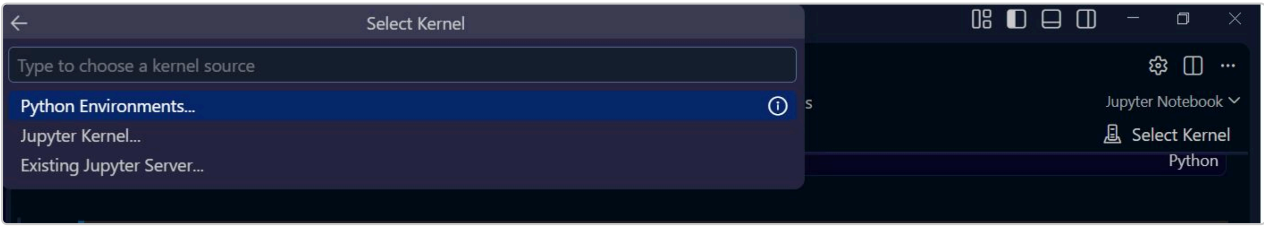
```
python -c "import numpy, pandas, matplotlib, scipy; print('all good')"
```

If it prints **all good**, your environment is ready and you are done. If it prints an error instead, see the next section.

Step 6. Choose the environment inside a notebook

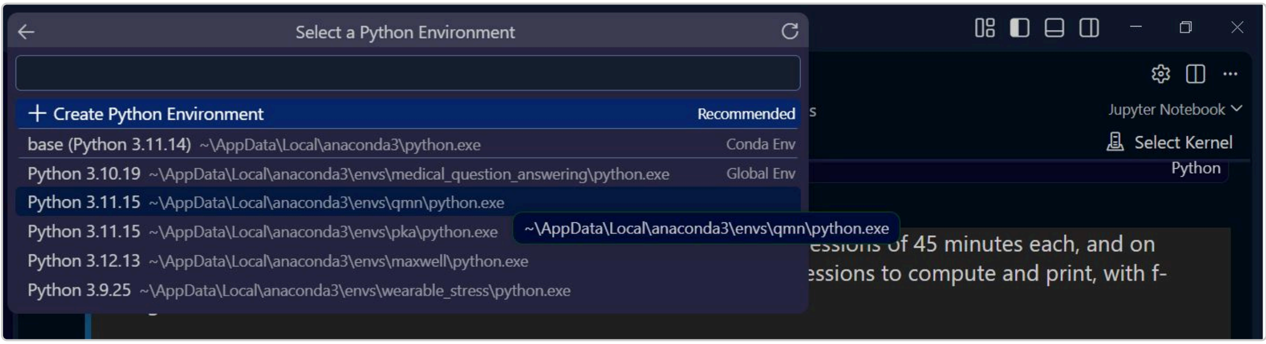
The last piece. A notebook has to be told which Python to use, and that choice is called the **kernel**. Open VS Code, use **File** then **Open Folder** to open your `QMN_practicals` folder, and open the first notebook by clicking it in the panel on the left.

Then click **Select Kernel**, at the top right of the notebook, and choose **Python Environments** from the panel that drops down:



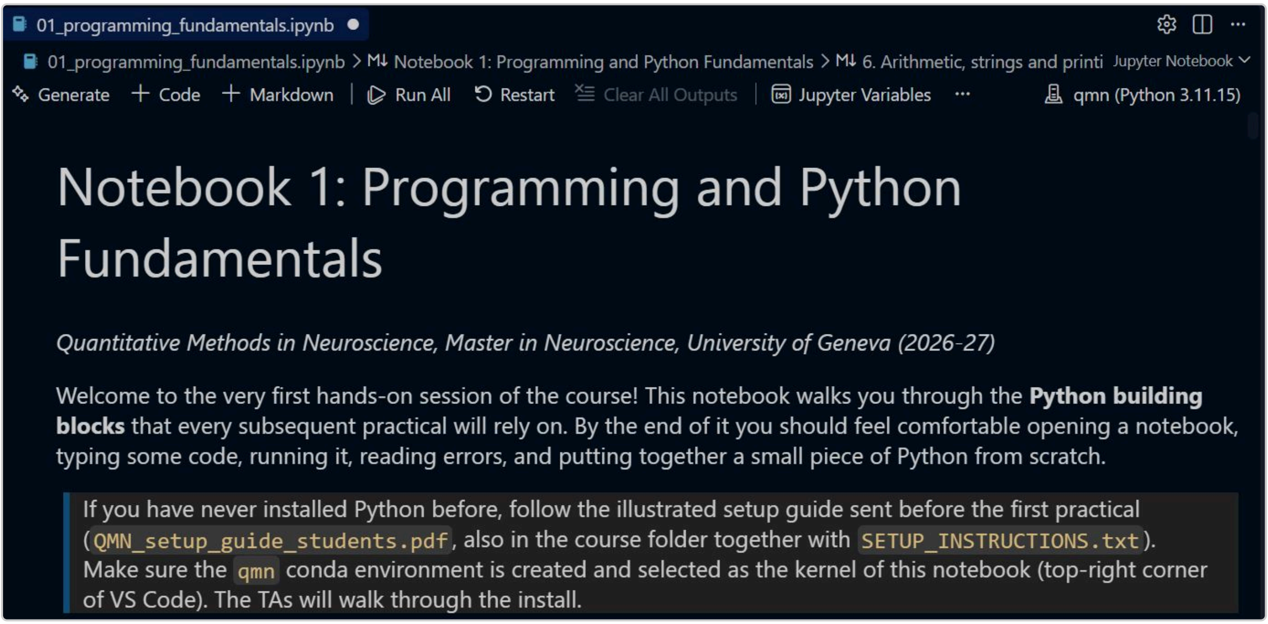
The button is at the top right. Of the three options offered, the one you want is Python Environments.

Now pick the `qmn` environment from the list:



The list identifies environments by their path, not by name, and the version number does not tell them apart. Look for the one whose path ends in `envs\qmn`. You will usually have only two, this one and `base`.

You will know it worked because the button in the top right stops saying "Select Kernel" and shows the environment instead:



Ready to work: the notebook is open, the kernel says qmn, and the toolbar has the Run All, Restart and Clear All Outputs buttons you will use all semester.

> We will do all of this together in the first practical, so if you got this far and something still does not look right, it is fine to stop and bring it with you.

If something goes wrong

Installation problems are normal and are usually specific to one machine. Two things help:

Ask an AI assistant, giving it this guide

Tools such as ChatGPT or Claude are genuinely good at this, and they help much more when they can see what you are following. Upload this PDF into the chat first, then copy the exact error message from the terminal, say which step you were on and which operating system you are using. For example: "Here is the setup guide I am following. I am on Windows 11, and at the step `conda env create -f environment.yml` I get this error, what should I do?" Then follow the suggestions one at a time, and check the result after each one before trying the next.

Write to us

If you are still stuck after twenty minutes, do not lose an evening over it. Send us an email at **giulio.matteucci@unige.ch** or **sami.el-boustani@unige.ch**, with your operating system, the step you reached, and a screenshot of the error, and we will sort it out with you. You can also come to the first practical with it unfinished: bring your laptop and we will fix it there.

Coming back to it later

You never repeat Step 4. Each time you work on the course, open the Anaconda Prompt or Terminal and run:

```
cd %USERPROFILE%\QMN_practicals
conda activate qmn
code .
```

The last line opens the course folder in VS Code.

One thing will happen during the semester: we may post an updated `environment.yml` when a later week needs a package that is not in it yet. Put the new file in the course folder, in place of the old one, and run this from there:

```
conda activate qmn
conda env update -n qmn -f environment.yml --prune
```

That brings the environment you already have up to date. You never create it a second time.
